# The Rental Penalty: Are Market-Rate Tenants More Exposed to Housing Cost Overburden in Europe?

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## 1 Project information

**Project title:** The Rental Penalty: Are Market-Rate Tenants More Exposed to Housing Cost Overburden in Europe?

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**GitHub repository:** https://github.com/davidvdbeek/format-assignment-

## 2 Introduction and research question

Housing affordability is a social problem because housing is a basic necessity and one of the largest fixed household costs. When housing costs take up a high share of disposable income, households may have less money left for food, transport, healthcare, education, energy and savings. Eurostat measures housing cost overburden as the share of the population living in households where total housing costs, net of housing allowances, are more than 40% of disposable household income (Eurostat, n.d.-a). Eurofound (2023) also argues that unaffordable housing can contribute to financial strain, housing insecurity and unequal living conditions.

This project focuses on a specific inequality inside the housing market: the difference between market-rate tenants and homeowners. This matters because renters and homeowners do not have the same economic position. Homeowners may build wealth through ownership, while market-rate tenants usually do not build housing wealth and may be more exposed to rent increases and rental market shortages. This makes market-rate tenants an important group when studying housing affordability.

**Main research question:**
**Do market-rate tenants face a higher housing cost burden than homeowners across European countries between 2014 and 2024?**

**Second-step analysis:**
After measuring this renter-homeowner difference, we examine whether countries with a larger rental penalty also have higher poverty or social exclusion risk. The at-risk-of-poverty-or-social-exclusion rate is used as a macro-level indicator of broader social vulnerability (Eurostat, n.d.-b). This does not test whether renting causes poverty. It only examines whether countries with a larger tenure-based housing burden gap also tend to have higher poverty/social exclusion risk.

## 2.1 Literature gap and contribution

Many analyses describe housing affordability as a general price-income problem. This project adds a more specific focus by looking at the distribution of housing cost burden between tenure groups. Instead of only asking whether housing costs are high, we ask whether market-rate tenants carry a systematically higher cost burden than homeowners with active housing payments. This shifts the focus from general affordability to **tenure-based inequality**.

The main new variable is the **rental penalty**:

$$Rental\ Penalty = Overburden\ rate\ of\ market\text{-}rate\ tenants - Overburden\ rate\ of\ owners\ with\ mortgage/loan$$

A positive value means that market-rate tenants are more likely to be overburdened by housing costs than owners with a mortgage or loan. We compare market-rate tenants with owners who still have a mortgage or loan because both groups usually face active monthly housing payments. Comparing tenants with owners without a mortgage or loan would likely exaggerate the gap, because many owners without a mortgage have much lower ongoing housing costs.

# 3 Data sourcing

We use two public Eurostat datasets. Both datasets are country-year datasets, which makes them suitable for comparing European countries over time. The housing dataset is used for the main analysis; the poverty/social exclusion dataset is used only for the second-step analysis.

| Dataset | Eurostat code | Unit of observation | Period | Key variables | Role in analysis | Main limitations |
|---|---|---|---|---|---|---|
| Housing cost overburden rate by tenure status | ilc_lvho07c | Country-year-tenure group | 2014-2024 selected | geo, year, tenure, values | Measures housing cost overburden and rental penalty | Country-level data; tenure categories may hide differences within countries |
| Persons at risk of poverty or social exclusion by age and sex | ilc_peps01n | Country-year-age-sex group | 2014-2024 selected | geo, year, age, sex, values | Measures broader poverty/social exclusion risk | Macro-level indicator; cannot identify individual household causes |

The variable `values` in the housing dataset is the housing cost overburden rate, expressed as a percentage. The variable `values` in the poverty dataset is the at-risk-of-poverty-or-social-exclusion rate, also expressed as a percentage. Because both variables are country-level percentages, the analysis can compare national patterns but cannot follow individual households.

# 4 Data cleaning and new variables

We restrict the analysis to 2014-2024 and percentage values. For the housing dataset, we keep the relevant tenure groups: owners with mortgage or loan (`OWN_L`), owners without mortgage or loan (`OWN_NL`), tenants at market rent (`RENT_MKT`), tenants at reduced/free rent (`RENT_FR`) and the total population (`TOTAL`). For the poverty dataset, we select total age and both sexes combined. The housing data are reshaped from long to wide format so that the rental penalty can be calculated for each country-year.

| Step | Dataset | Operation | Observations after | Rationale |
|---|---|---|---|---|
| Raw housing data | Housing | Downloaded from Eurostat | 3925 | Starting point |
| Filter years 2014-2024 and percentage unit | Housing | Keep TIME_PERIOD 2014-2024 and unit == PC | 2220 | Comparable period and unit |
| Select tenure groups and valid country names | Housing | Keep OWN_L, OWN_NL, RENT_MKT, RENT_FR, TOTAL | 1905 | Needed to compare renters and homeowners |
| Reshape housing data to wide format | Housing | One row per country-year with tenure columns | 381 | Needed to calculate rental penalty |

| | | | | |
|---|---|---|---|---|
| Join with poverty/social exclusion data | Combined | Match country-year observations | 345 | Needed for second-step poverty analysis |

These cleaning choices involve trade-offs. Country-level data allow broad European comparison but cannot describe individual households. Tenure categories may also differ across national housing systems. Therefore, the renter-homeowner difference should be interpreted as descriptive, not causal.

| New variable | Formula / construction | Analytical purpose |
|---|---|---|
| rental_penalty | RENT_MKT - OWN_L | Measures the renter-homeowner burden gap |
| post_2020 | year >= 2020 | Separates the event period |
| rental_penalty_change | average post-2020 rental penalty - average pre-2020 rental penalty | Measures whether the gap changed after 2020 |
| high_penalty_country | above or below median rental penalty | Groups countries for interpretation |

The average rental penalty in the cleaned sample is 22.3 percentage points across 36 countries. This means that, on average, market-rate tenants have a substantially higher overburden rate than owners with a mortgage or loan.

# 5 Results

## 5.1 Sub-population variation: housing burden by tenure group

Tenure status is treated as the sub-population dimension because it separates households by their position in the housing market. Market-rate tenants are especially important because they are directly exposed to rental market prices. Owners with a mortgage or loan are the main comparison group because they also usually face active monthly housing payments.

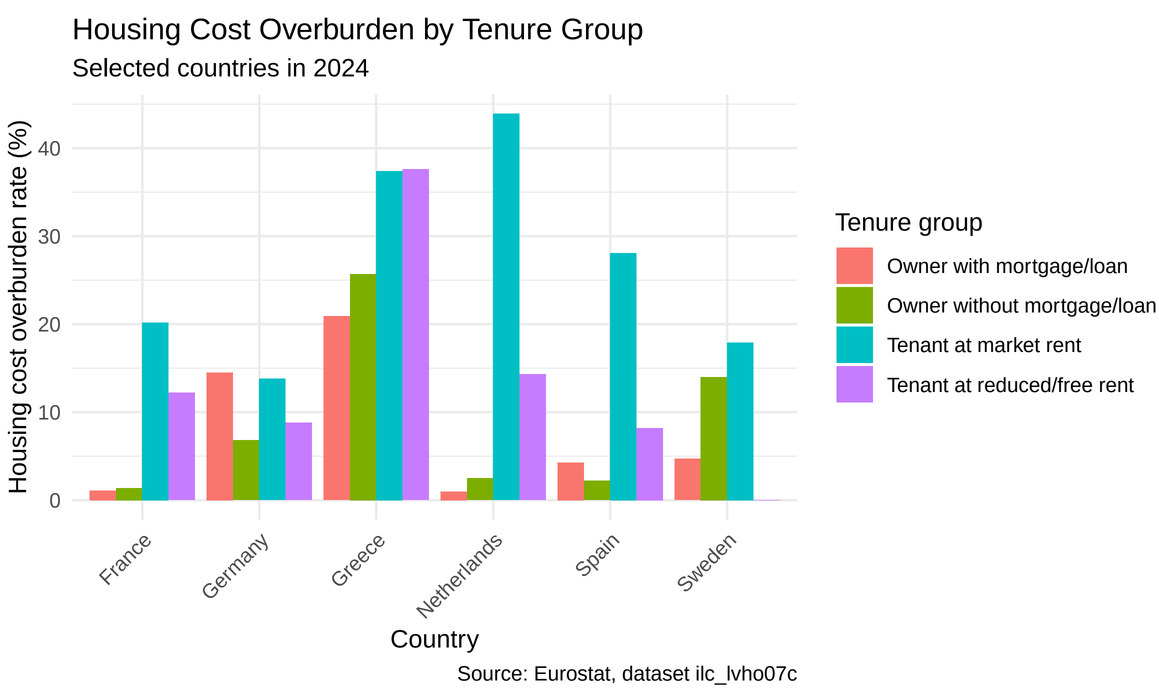


Figure 1: Housing cost overburden by tenure group in selected countries.

Figure 1 shows that market-rate tenants usually face a higher overburden rate than owners with a mortgage. In the Netherlands, for example, the overburden rate is 43.9% for market-rate tenants compared with 1.0% for owners with a mortgage. Spain and France also show clear tenure-based gaps. Greece is different: several tenure groups show high overburden rates, which suggests a broader affordability problem rather than only a rental-market problem.

## 5.2 Spatial variation: where is the rental penalty largest?

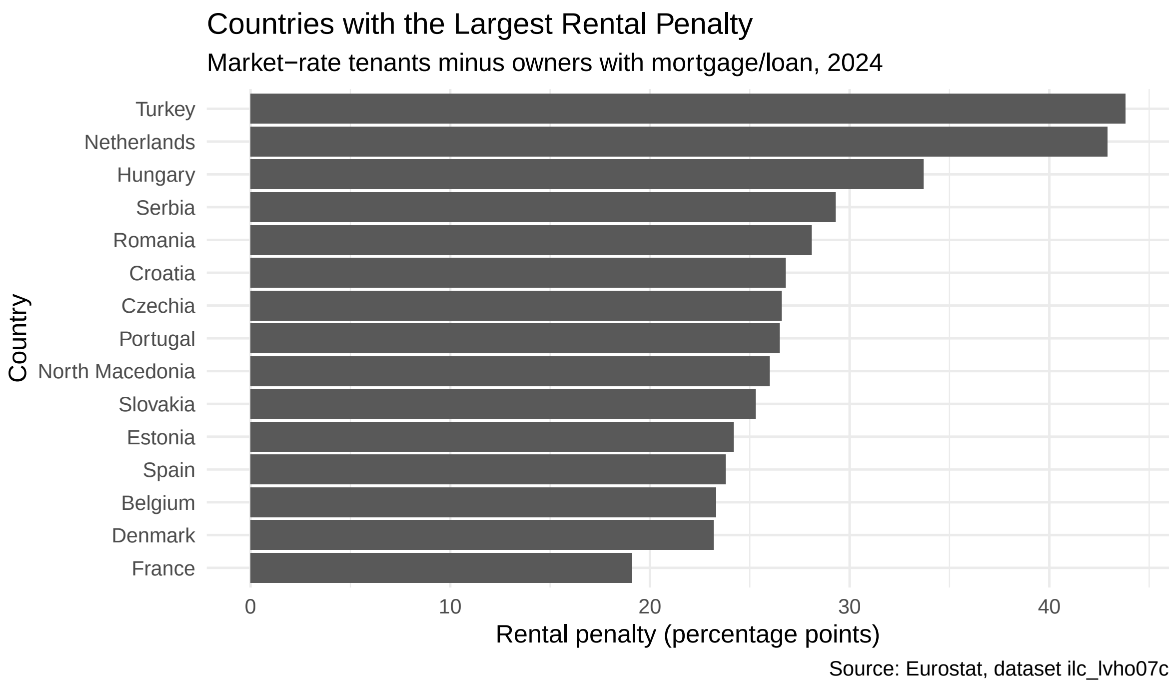


Figure 2: Countries with the largest rental penalty in the latest available year.

A high rental penalty does not necessarily mean that all housing costs are highest in that country. It means that the difference between market-rate tenants and homeowners with a mortgage or loan is largest. In the latest available year, the largest penalties are found in countries such as Turkey, the Netherlands and Hungary. Countries with low or missing values should also be interpreted cautiously, because country-level comparisons can hide regional and urban-rural differences.

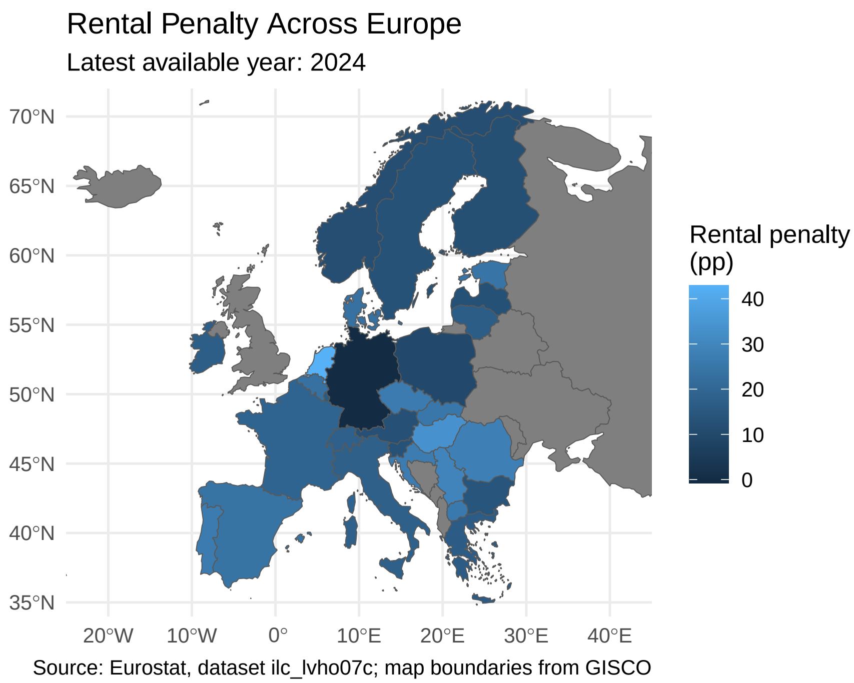


Figure 3: Spatial distribution of the rental penalty across Europe.

The map shows country-level spatial differences in the rental penalty. It is useful for seeing broad European patterns, but it has limits: small countries can be visually underemphasised, missing data appear as blank areas, and the map does not show within-country differences such as city versus rural housing markets.

### 5.3 Temporal variation: did the rental penalty change over time?

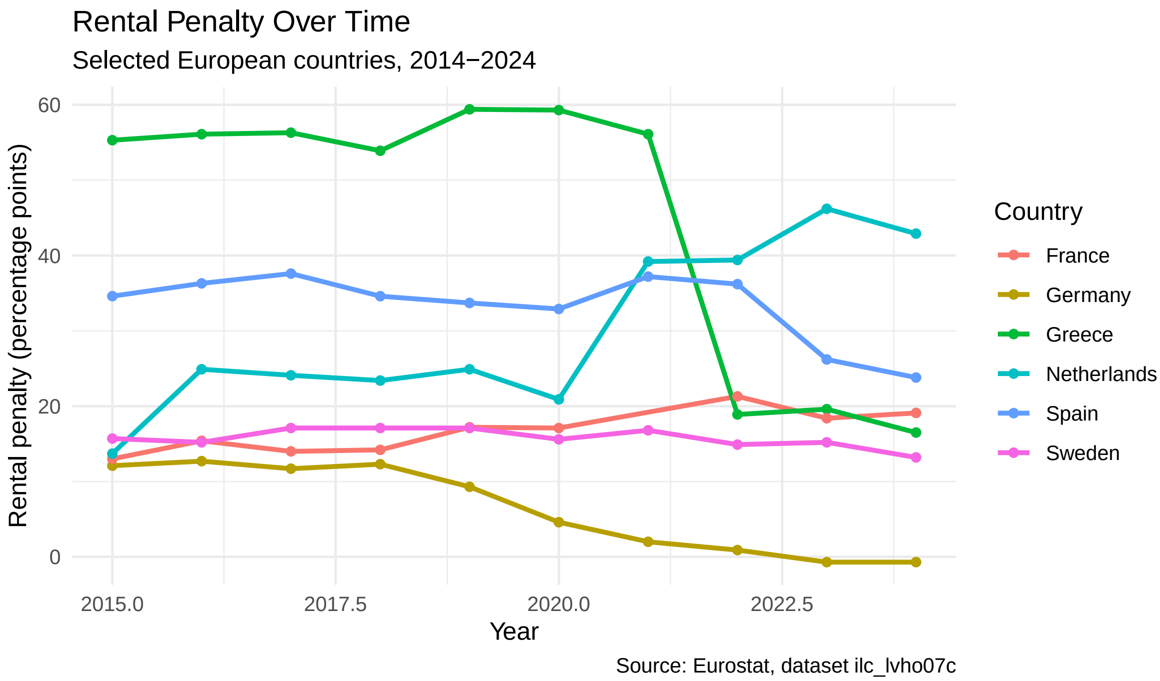


Figure 4: Rental penalty over time in selected countries.

| country | first_year | first_penalty | last_year | last_penalty | change | trend |
|---|---|---|---|---|---|---|
| France | 2015 | 13.0 | 2024 | 19.1 | 6.1 | Increasing |
| Germany | 2015 | 12.1 | 2024 | -0.7 | -12.8 | Decreasing |
| Greece | 2015 | 55.3 | 2024 | 16.5 | -38.8 | Decreasing |
| Netherlands | 2015 | 13.7 | 2024 | 42.9 | 29.2 | Increasing |
| Spain | 2015 | 34.6 | 2024 | 23.8 | -10.8 | Decreasing |
| Sweden | 2015 | 15.7 | 2024 | 13.2 | -2.5 | Relatively stable |

The temporal pattern suggests that the rental penalty is not a uniform European trend. Greece shows a strong decline after 2021, while the Netherlands shows a large increase compared with the start of the period. Spain remains high for much of the period, while Sweden is more stable. These differences may reflect national variation in rental regulation, mortgage exposure, housing supply constraints, household income growth and policy responses. A trend in this graph does not prove a policy or COVID effect.

### 5.4 Event association: did the rental penalty change after 2020?

We use 2020 as a descriptive event point because it marks the start of the COVID-19 period and is followed by wider cost-of-living pressure in Europe. This is not a causal test. The pre/post comparison does not control for national policy responses, inflation, housing supply, mortgage interest rates or income support measures.

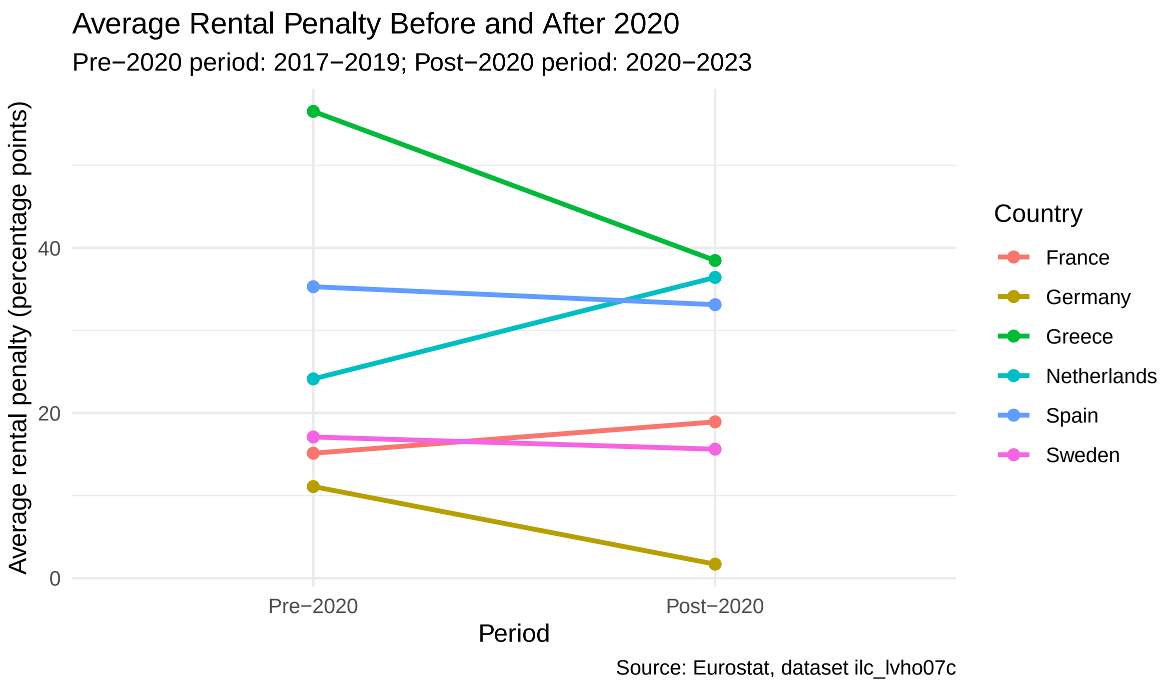


Figure 5: Average rental penalty before and after 2020.

Figure 5 shows mixed patterns in the pre/post comparison. The Netherlands increases after 2020, while Greece and Germany decrease. This supports the idea that there is no single European post-2020 effect. The results show whether the rental penalty changed around the post-2020 period, but they do not prove that COVID-19 or the cost-of-living crisis caused the change.

### 5.5 Second-step analysis: rental penalty and poverty risk

The second-step analysis asks whether countries with a larger rental penalty also show higher poverty or social exclusion risk. Both variables are measured at country-year level. This is a macro-level association and does not show that renting causes poverty. Possible confounders include welfare state design, income inequality, housing policy, rent regulation, labour market structure and household composition.

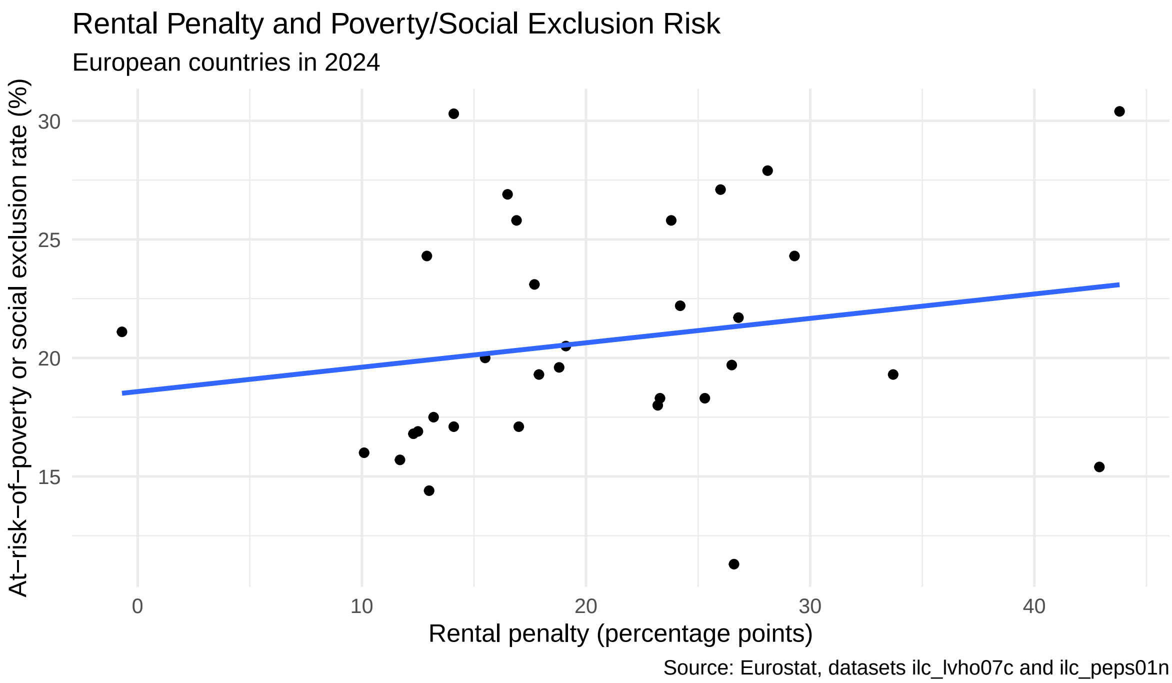


Figure 6: Rental penalty and poverty/social exclusion risk in the latest available year.

| Latest year | Correlation between rental penalty and poverty/social exclusion risk |
|---|---|
| 2024 | 0.2 |

The relationship is positive, with a correlation of approximately 0.2. This suggests that countries with a larger renter-homeowner burden gap may also have higher poverty/social exclusion risk, but the scatterplot shows substantial variation. Therefore, rental market inequality and poverty risk overlap only partly. Some countries may have a large rental penalty without the highest poverty risk, which suggests that housing market structure can matter independently from general poverty levels.

# 6 Discussion and conclusion

The analysis shows that market-rate tenants generally face a higher housing cost overburden rate than owners with a mortgage or loan. This answers the main research question: in many European countries, there is a measurable rental penalty. Housing affordability is therefore not only a general price-income problem, but also a distributional issue between tenure groups.

The size of the rental penalty differs strongly across Europe. Countries such as Turkey, the Netherlands and Hungary show large renter-homeowner gaps in the latest available year, while other countries have smaller gaps. This implies that housing affordability problems cannot be understood only by looking at average national housing costs. It matters who carries the burden.

The time-series and event analyses show that the rental penalty does not develop in the same way everywhere. Some countries show persistent or increasing gaps, while others show declining gaps after 2020. These differences may reflect national housing systems, rent regulation, mortgage conditions, housing supply and income support policies. The variation in post-2020 trajectories may also reflect differences in national policy responses to the pandemic, as countries differed substantially in the type and scope of housing protections they introduced for both tenants and mortgaged homeowners (Krapp et al., 2020). However, the analysis is descriptive and cannot prove which factor caused the changes.

The second-step analysis suggests that countries with larger rental penalties may also have higher poverty/social exclusion risk, but the relationship is not perfect. This means that housing tenure inequality may be part of broader social inequality, while also being shaped by housing-market institutions and policy choices that differ between countries.

The main limitation is that the data are country-level indicators. They cannot compare individual households with the same income, family size or location. Future research could use household-level microdata, regional data or housing policy indicators to test whether renters remain more exposed after controlling for income, household composition and urban housing markets.

# 7 Reproducibility

## 7.1 GitHub repository link

The knitted PDF contains the link to the public GitHub repository for this project:

https://github.com/davidvdbeek/format-assignment-

## 7.2 How this project is reproduced

The project can be reproduced from the files in the GitHub repository. The repository contains the RMarkdown file used to generate this report, a README file explaining the workflow, and the required reproducibility files.

```
rental_penalty_project/
  README.md                         # explains the project and how to run it
  rental-penalty-final.Rmd    # this report
  rental-penalty-final.pdf    # knitted PDF report
  renv.lock                         # package environment
```

# 8 References


Eurofound. (2023). *Unaffordable and inadequate housing in Europe*. Publications Office of the European Union.

Eurostat. (n.d.-a). *Glossary: Housing cost overburden rate*. Statistics Explained.

Eurostat. (n.d.-b). *Glossary: At risk of poverty or social exclusion (AROPE)*. Statistics Explained.

Eurostat GISCO. (n.d.). *Countries - administrative units*. European Commission.

Eurostat. (2026). *Housing cost overburden rate by tenure status* [Data set]. Dataset code: `ilc_lvho07c`.

Eurostat. (2026). *Persons at risk of poverty or social exclusion by age and sex* [Data set]. Dataset code: `ilc_peps01n`.

OECD. (2024). *HC1.2 Housing costs over income*. OECD Affordable Housing Database.

Krapp, M. C., Vaché, M., Egner, B., Schulze, K., & Thomas, S. (2020). *Housing policies in the European Union*. Institut Wohnen und Umwelt GmbH: Darmstadt, Germany.